SETP4583

SURFACTANT AND DETERGENT TECHNOLOGY

SECTION EB01

INDIVIDUAL ASSIGNMENT:

Individual Reflective Journal Assignment

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1.0 Introduction

Figure 1.1 Our Class Photo

When I choose to enrol in the Surfactant and Detergent Technology course, I assumed it would be a purely theoretical chemistry class filled with complex formulas and chemical bonds, much like the Chemistry subject I studied in my secondary school. As a Computer Science student, I specifically chose this course because I wanted a break from working with computers and was eager to learn something different that is closely related to our daily lives. My main goal was to understand the actual science behind the products I use every day. While I expected to spend most of my time understanding the difficult chemical structures, I was pleasantly surprised to find that the course focused more on deep understanding, interactive presentations, sustainability, and practical problem-solving in the project experiment. This journal reflects my rewarding journey from learning the basics of surfactants to successfully leading a group project to create an eco-friendly body wash. It is really a nice experience for me that I found it incredibly satisfying.

2.0 Reflection on Key Learnings

The most important concept I learned throughout this course was the synergistic nature of surfactants which describes how different cleaning agents work together to achieve a specific goal. Before this class, I simply thought soap was one single ingredient but now I understand that creating an effective cleanser requires a balanced system and very precise measurements of various surfactants. This realization changed my perspective on product design entirely. I learned that for a detergent to be successful, it must balance cleaning power, foam stability, and skin mildness.

In our group project, I learned the specific roles of different surfactant types. For instance, Sodium Laureth Sulfate (SLES) acted as our primary anionic surfactant because of its high foaming capacity and its effectiveness in removing oils and grease. In our final formulation, we used a concentration of 12% SLES to ensure strong detergency. I also learned about Linear Alkylbenzene Sulfonate (LABS), which is a powerful secondary anionic surfactant used for grease-cutting. However, a key lesson was that LABS can be very harsh on the skin if used in high amounts. Therefore, we chose to use it at a very low percentage, only 1% to strengthen the cleaning performance without making the product too aggressive.

My "aha" moment occurred when I learned about Betaine, an amphoteric surfactant. I realized that by adding 8% Betaine to our mix, we could significantly reduce the harshness of the anionic surfactants and make the foam much more stable and creamier. This taught me that chemistry is about finding the right "partners" for each chemical to make them work effectively and safely. Furthermore, learning about non-ionic surfactants like Berol expanded my understanding even more. I understand how non-ionic surfactants help the formula work effectively regardless of water hardness or pH differences, which is vital for a product that might be used in different regions with different water types.

The laboratory sessions and the group project were the main point of my learning experience. In the lab, we did not just learn about surfactants, we saw how they interacted as we mixed them together to create a detergent mixture. By creating our product, PURELUX Gentle Body Wash, it taught me that a formulation is more than just a list of ingredients. It is about finding the accurate ratios of each surfactant to achieve the perfect balance. During our early trials, we faced a major challenge when we used bottle caps to estimate our measurements which resulted in an undesirable slime-like texture. This experience turned abstract concepts into tangible skills by showing me that even a small error in measurement can completely change the texture and performance of the final product. By changing into using syringes for accuracy and mixing at room temperature, I learned the importance of precision and sustainability in real-world science.

3.0 Application and Real-World Relevance

There are many ways to apply the knowledge I gained from this course in a real-world context. One of the most important connections I made to industry practices was the use of the "cold mix" process. In traditional manufacturing, many companies use a large amount of heat to mix ingredients, which consumes a lot of electricity. By learning to mix our body wash at room temperature, I understood how industries can reduce their carbon footprint and save money. In the future, I can apply this mindset of process optimization to any field, looking for other ways to make production more energy-efficient and eco-friendly.

I also discovered how closely industry standards are linked to human biology and environmental safety. For example, the requirement for a detergent to be pH-balanced between 5.5 and 6.0 is not just a random number. It is a professional standard designed to mimic the natural protective barrier of human skin. Understanding this helps me realize that real-world products must be designed with the end user's health in mind. Additionally, the usage of biodegradable surfactants like SLES and Betaine is a major trend in the modern chemical industry. These ingredients break down quickly in water which prevents them from harming fish or polluting our waterways.

Finally, I learned about the importance of sustainable packaging and clear labelling in the real world. Our project recommended using a refillable pump-bottle design which is a practice in the industry to help customers reduce plastic waste. I also learned that professional products must have clear safety instructions such as warnings to keep the soap away from eyes and stating that it is for external use only. This knowledge shows me that a successful product is not just about the chemistry inside the bottle but also about how it is packaged, labelled and disposed of to ensure it is safe for both the consumer and the planet.

4.0 Challenges and Growth

The most significant challenge I faced throughout this course was overcoming the gap in my chemistry knowledge. Although I was strong in chemistry during secondary school, I found that I had forgotten many fundamental concepts. During lectures, I struggled to recall basics such as the behaviours of anions and cations, the principles of surface tension and maybe the specific terms like hydrophobic and hydrophilic. Since our course involved many presentations and class discussions on specific topics, I am struggling to perform well. To overcome this, I had to spend extra time doing research to relearn these concepts from scratch so that I could contribute meaningfully to group activities and present our findings with confidence. This process taught me that learning is a continuous journey and even forgotten skills can be recovered with enough effort and research.

Figure 4.1Slime-like Texture Failed Result

Beyond the academic struggle, our team faced practical challenges during the body wash product formulation phase. At first, we lacked of professional measuring tools which forced us to estimate surfactant amounts using bottle caps. This lack of precision led to our first major failure; the mixture developed a "slime-like" texture instead of becoming a smooth body wash. We addressed this by switching to syringes which allowed for much more accurate measurements and finally led to a successful product. This experience taught me that in both chemistry and data engineering, precision and accuracy is everything. Even a small error in a "code" or a "formula" can lead to a failed result and that having the right tools and a research-based mindset is the key to solving any problem.

5.0 Personal and Professional Development

This course has significantly influenced my professional mindset by bridging the gap between my background in Computer Science and the physical world of product engineering. I have developed a deep appreciation for precision, just as one small error in a line of code can stop a system from running successfully, a slightly error in a surfactant ratio can turn a body wash into a failed "slime". This experience has reinforced my commitment to accuracy in all my technical work in the future whether it is digital or physical. Furthermore, a major learning outcome for me was the importance of problem-solving. I realized that whether you are debugging code or conducting chemical experiments, there will always be challenges that require you to think out of the box and always willing to find solutions.

My skillset has also expanded to include better research, try and test and group collaboration. As a result of this course, I am eager to explore how I can use my data engineering skills to optimize detergent science. I am particularly interested in using algorithms to predict the best substances for specific needs. For example, calculating formulas that are scientifically proven to be extra gentle for babies. On a more practical level, I want to further investigate how to add moisturizing agents like glycerin or natural extracts such as aloe vera to improve skin hydration. This was a limitation we identified in our own prototype after seeing other group presentations. Ultimately, this course has shown me that science is incredibly satisfying when it is applied to solving real-world and daily-life problems.

Figure 5.1 Our Group Photo